

USB4 2.0 ENGINEERING CHANGE NOTICE FORM

Title: Clarification on sleep entry flow for a device router operating in TBT3 mode – V3

Applied to: USB4 Specification Version 2.0

Brief description of the functional changes:

Changing the description of sleep entry flow described in Chapter 13 to clearly list down the expected behavior and differences when a USB4 device router operates in TBT3 mode.
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Benefits as a result of the changes:

Making the required behavior of a USB4 device Router that supports TBT3 clearer when working with a TBT3 CM.
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An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
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None

An analysis of the hardware implications:
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None

An analysis of the software implications:
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None

An analysis of the compliance testing implications:
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None

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Actual Change

(a). 13.2.4 Sleep and Wake

To Text:

This section only applies to a Device Router that supports TBT3-Compatibility on its Upstream Facing Port. If bits 15:12 in the Connection Manager *USB4 Version* field in Router Configuration Space are 0b (indicating a TBT3 Connection Manager), a Router shall support sleep and wake per Section ~~4.4.7~~ 4.5 with the changes defined in this section.

(b). 13.2.4.1 - Entry to Sleep

To Text:

~~After the Enter Sleep bit is set to 1b in all Ports, a Device Router shall do the following for each USB4 Port:~~

- ~~• Transition the Lane Adapters to CLd state.~~
- ~~• If any of the following conditions apply, the USB4 Port shall go through disconnect:~~
 - ~~○ For Lane 0 in a USB4 Port:~~
 - ~~▪ The Lane 0 is Inter-Domain bit is 0b and the Lane 0 Configured bit is 0b.~~
 - ~~▪ The Lane 0 is Inter-Domain bit is 1b and the Inter-Domain Disconnect on Sleep bit is set to 1b.~~

After the Enter Sleep bit is set to 1b in all Ports, a Device Router ~~DR~~operating with a TBT3 Connection Manager, shall follow the steps mentioned in Section 4.5.1 with the following differences:

- Lane 0 is Inter-Domain bit will be checked instead of USB4 Port is Inter-Domain bit.
- Lane 0 Configured bit will be checked instead of USB4 Port is Configured bit.
- Inter-Domain Disconnect on Sleep bit = 1b will be checked instead of Enable Wake on Inter-Domain bit = 0b.